MODULE 34 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 4 of 4

Twenty new questions, weighted like the real exam, including multiple-response questions.

QUESTIONS 20 PASS MARK 15 of 20 HOLD 9 seconds

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How to use this paper

- Pause after each question and commit to an answer before the solution appears.
- Write down your letters. Multiple-response questions need every correct letter.
- Score yourself by **domain** at the end, not only in total.
- For each miss, revisit the module named on the solution slide.

> ON THE EXAM
>
> The answer teaches the fact. The module teaches the discriminator you missed.

> A V2 pipeline validates pull requests. When several are opened together, each waits for the one before it, and feedback takes an hour.
>
> Each pull request deploys to its own isolated test stack, and runs share nothing.

What should the DevOps engineer change?

A Set the execution mode to QUEUED

B Set the execution mode to SUPERSEDED

C Set the execution mode to PARALLEL

D Add more CodeBuild compute

◆ PAUSE AND DECIDE

What should the DevOps engineer change?

A Set the execution mode to QUEUED

Queued is what makes them wait in line.

B Set the execution mode to SUPERSEDED

Newer runs would replace older ones, skipping pull requests.

C Set the execution mode to PARALLEL

Isolated runs proceed independently, so every pull request gets fast feedback.

D Add more CodeBuild compute

The runs are waiting in line, not short of compute.

REVISIT MODULE 2

Isolated runs, fast independent feedback: PARALLEL. Order matters: QUEUED.

> An ECS service uses CodeDeploy blue/green with a test listener on port 8443. Integration tests must run against the green tasks before any production traffic.
>
> If the tests fail, the deployment must roll back automatically.

Where should the tests run?

A In a Lambda function on the AfterAllowTestTraffic hook, reporting failure to CodeDeploy

B In the AfterAllowTraffic hook

C In a CodeBuild stage after the deployment completes

D In a BeforeInstall hook

◆ PAUSE AND DECIDE

Where should the tests run?

A In a Lambda function on the AfterAllowTestTraffic hook, reporting failure to CodeDeploy

Runs once green is reachable on the test listener; a failure rolls back.

B In the AfterAllowTraffic hook

Production traffic has already moved by then.

C In a CodeBuild stage after the deployment completes

Customers already reach green, and CodeDeploy has finished.

D In a BeforeInstall hook

Green is not running or reachable yet.

REVISIT MODULE 7

Test green on the test listener, before production: AfterAllowTestTraffic.

> Builds pull public base images from Docker Hub and hit its rate limits. Security also wants those images scanned inside AWS.
>
> The team wants the least operational overhead.

What should the DevOps engineer set up?

A. A nightly job copying every needed image into ECR

B. A NAT gateway with more IP addresses

C. Build base images from source in every build

D. An ECR pull through cache rule for Docker Hub, with credentials in Secrets Manager

◆ PAUSE AND DECIDE

What should the DevOps engineer set up?

A A nightly job copying every needed image into ECR
Custom, and misses newly needed images.

B A NAT gateway with more IP addresses
Rate limits follow the account, and nothing is scanned.

C Build base images from source in every build
Slow and wasteful.

D An ECR pull through cache rule for Docker Hub, with credentials in Secrets Manager
Images are cached in ECR after the first pull, and ECR scanning applies.

REVISIT MODULE 5

Public registry limits, and scanning in AWS: pull through cache.

> A buildspec sets a database password as a plaintext environment variable, visible to anyone who can view the project.
>
> The password already lives in Secrets Manager, encrypted with a customer managed key.

Which steps fix this? (Choose TWO.)

A Encrypt the buildspec file in the repository

B Echo the value in the build log to confirm it works

C Reference the secret in the buildspec's secrets-manager section

D Move the value into a project environment variable of type PLAINTEXT

E Allow secretsmanager:GetSecretValue and kms:Decrypt on the service role

◆ PAUSE AND DECIDE

Which steps fix this? (Choose TWO.)

A Encrypt the buildspec file in the repository

The build needs to read it; the value is still plain.

B Echo the value in the build log to confirm it works

Exposes it further.

C Reference the secret in the buildspec's secrets-manager section

Resolved at build time; masked in logs; never stored in the project.

D Move the value into a project environment variable of type PLAINTEXT

Still visible to anyone who can view the project.

E Allow secretsmanager:GetSecretValue and kms:Decrypt on the service role

The build runs as the service role, which must be able to fetch and decrypt.

REVISIT MODULE 3

Reference, do not store. And the service role, not the developer, must be able to read it.

A StackSet deploys a template to 30 accounts in two Regions. It succeeds in the first Region, then fails in the second with 'already exists' errors.

The template gives an IAM role and an S3 bucket fixed names.

What should the DevOps engineer change?

A Increase the failure tolerance

B Include the Region in the names, or let CloudFormation generate them

C Deploy the Regions in parallel

D Switch to self-managed permissions

◆ PAUSE AND DECIDE

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What should the DevOps engineer change?

A Increase the failure tolerance

Lets more stacks fail; they still collide.

B Include the Region in the names, or let CloudFormation generate them

IAM names are account-wide and bucket names global; unique names stop the collision.

C Deploy the Regions in parallel

Collides sooner.

D Switch to self-managed permissions

Permissions are not the problem.

REVISIT MODULE 10

Second Region fails with 'already exists': a globally unique name is repeated.

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> A central team maintains the approved AMI ID in a Parameter Store parameter in a shared services account.
>
> Launch templates in 40 workload accounts must read that one parameter directly, always getting the current value.

What should the DevOps engineer do?

A Copy the parameter into each account with a scheduled Lambda

B Store the value in a public S3 object

C Make it an advanced-tier parameter and share it through AWS RAM

D Move the value into Secrets Manager with a resource policy

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Copy the parameter into each account with a scheduled Lambda

Copies can drift; not the one parameter.

B Store the value in a public S3 object

Public, and launch templates cannot reference it.

C Make it an advanced-tier parameter and share it through AWS RAM

Other accounts read the one shared parameter directly.

D Move the value into Secrets Manager with a resource policy

Not a secret, and launch templates resolve parameters, not secrets.

REVISIT MODULE 25

One parameter read by many accounts: advanced tier, shared with RAM.

> A service-managed StackSet targets the organisation root with a security baseline. Every member account received it.
>
> The management account did not, and auditors require the baseline there too.

What should the DevOps engineer do?

A Deploy the template to the management account as a separate stack

B Retry the StackSet operation

C Move the management account into an OU

D Increase maximum concurrent accounts

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Deploy the template to the management account as a separate stack

Service-managed StackSets never deploy to the management account, by design.

B Retry the StackSet operation

It will skip the management account again.

C Move the management account into an OU

It is still the management account, and still skipped.

D Increase maximum concurrent accounts

Speed is not the problem.

REVISIT MODULE 10

Missing only in the management account: expected behaviour. Deploy there separately.

> A company with 70 accounts needs daily backups of all resources tagged backup=daily, kept 35 days, and copied to a central vault in a separate backup account.
>
> New accounts must be covered automatically.

What should the DevOps engineer use?

A. A backup plan created by hand in each account

B. A Lambda function in each account taking snapshots

C. Data Lifecycle Manager policies for EBS only

D. An AWS Organizations backup policy with a cross-account copy rule to the central vault

◆ PAUSE AND DECIDE

What should the DevOps engineer use?

A. A backup plan created by hand in each account
Seventy set-ups, and new accounts are missed.

B. A Lambda function in each account taking snapshots
Custom, per service, and per account.

C. Data Lifecycle Manager policies for EBS only
EBS only, and no cross-account vault copy.

D. An AWS Organizations backup policy with a cross-account copy rule to the central vault
Applied to every account by tag, including new ones, with copies to the central vault.

REVISIT MODULE 16

Same backup plan everywhere, including new accounts, with central copies: an organisation backup policy.

> An application on Amazon RDS for PostgreSQL Multi-AZ suffers several minutes of errors after each failover, although the failover itself completes in about a minute.
>
> Investigation shows the application keeps using stale connections and cached DNS.

Which steps reduce the impact of failover? (Choose TWO.)

A Put RDS Proxy in front of the database

B Move to a larger instance class

C Disable Multi-AZ

D Use a short DNS cache time, and retry with backoff on connection errors

E Add a read replica

◆ PAUSE AND DECIDE

Which steps reduce the impact of failover? (Choose TWO.)

A Put RDS Proxy in front of the database

Clients stay connected to the proxy while it reconnects to the new primary.

B Move to a larger instance class

Size does not affect stale connections.

C Disable Multi-AZ

Removes automatic failover altogether.

D Use a short DNS cache time, and retry with backoff on connection errors

Clients find the new primary quickly and recover without a stampede.

E Add a read replica

Serves reads; does not help writers reconnect.

REVISIT MODULE 15

The database recovered in a minute; the clients took minutes. Fix the clients' path back.

> A team ran an AWS FIS experiment that stopped instances in one Availability Zone. Customer errors spiked, but the experiment kept running for its full 30 minutes.
>
> Future experiments must halt automatically if customers are affected.

What should the DevOps engineer add?

A A shorter experiment duration

B A stop condition using a CloudWatch alarm on the customer error rate

C A manual approval before each experiment

D Run experiments only at night

◆ PAUSE AND DECIDE

What should the DevOps engineer add?

A. A shorter experiment duration
Still runs regardless of harm, just for less time.

B. A stop condition using a CloudWatch alarm on the customer error rate
Halts the experiment as soon as customers are affected.

C. A manual approval before each experiment
Approves the start; stops nothing mid-run.

D. Run experiments only at night
Fewer customers, but still no automatic halt.

REVISIT MODULE 16

Halt on harm: an FIS stop condition tied to an alarm.

> An API's latency alarm uses the Average statistic and never fires. Yet a steady share of customers complain that requests take several seconds.
>
> Most requests are fast, so the average stays low.

What should the DevOps engineer change?

A Alarm on the p99 latency percentile

B Lower the average threshold

C Use the Sum statistic

D Use a longer period

◆ PAUSE AND DECIDE

What should the DevOps engineer change?

A **Alarm on the p99 latency percentile**
Reflects the slowest requests that averages hide.

B **Lower the average threshold**
Starts firing on normal traffic while still blurring the slow tail.

C **Use the Sum statistic**
Sums latency, which grows with traffic, not slowness.

D **Use a longer period**
Smooths even more of the slow tail away.

REVISIT MODULE 18

Averages hide the tail. Alarm on a high percentile for user experience.

> Containers on ECS with Fargate must send logs to CloudWatch Logs and, filtered to errors only, to a third-party log platform.
>
> The team wants a managed pattern without changing application code.

What should the DevOps engineer use?

A The awslogs driver alone

B A metric stream to the partner

C FireLens with Fluent Bit routing to both destinations, filtering the partner output

D Have the application write to both platforms

◆ PAUSE AND DECIDE

What should the DevOps engineer use?

A The awslogs driver alone

Sends to CloudWatch only; no second destination or filtering.

B A metric stream to the partner

Carries metrics, not logs.

C FireLens with Fluent Bit routing to both destinations, filtering the partner output

A sidecar router with multiple outputs and filters, configured in the task definition.

D Have the application write to both platforms

A code change, which is ruled out.

REVISIT MODULE 17

Several log destinations, with filtering, from containers: FireLens.

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A company needs every application log from 50 accounts kept for five years in a central account. Operators also want to search recent logs across all accounts from a monitoring account, in the same Region.

It wants the least operational overhead.

Which steps meet these requirements? (Choose TWO.)

A Nightly export tasks from every log group

B CloudWatch Logs centralisation rules copying to the central account

C Cross-account observability links to the monitoring account

D A metric stream from every account

E Subscription filters on every log group, streaming to Kinesis

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose TWO.)

A Nightly export tasks from every log group
Batch jobs to schedule everywhere.

B CloudWatch Logs centralisation rules copying to the central account
The retained copy, organisation-wide, with no code.

C Cross-account observability links to the monitoring account
Operators search logs in place, without copying.

D A metric stream from every account
Metrics, not logs.

E Subscription filters on every log group, streaming to Kinesis
Works, with far more to build and run.

REVISIT MODULE 17

Retain: copy with centralisation. Search in place: cross-account observability. Two needs, two tools.

> Workloads run in us-west-2 and eu-west-1. The central operations team's automation listens on an event bus in us-east-1.
>
> GuardDuty findings from both Regions must reach that bus with the least effort.

What should the DevOps engineer set up?

A. A Lambda function in each Region that re-publishes findings

B. Move all workloads to us-east-1

C. Poll GuardDuty in each Region every minute

D. In each Region, a rule on GuardDuty findings targeting the us-east-1 event bus

◆ PAUSE AND DECIDE

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What should the DevOps engineer set up?

A A Lambda function in each Region that re-publishes findings

Custom code for a native feature.

B Move all workloads to us-east-1

Absurd scope for routing events.

C Poll GuardDuty in each Region every minute

Polling instead of events.

D In each Region, a rule on GuardDuty findings targeting the us-east-1 event bus

EventBridge can send events to a bus in another Region directly.

REVISIT MODULE 21

Events from many Regions to one bus: a rule in each Region targeting that bus.

> A remediation must snapshot an EBS volume, wait until the snapshot completes, which can take hours, then detach and delete the volume.
>
> It must be triggered automatically from a Config rule and retry failed steps.

What should implement the remediation?

A. A single Lambda function that waits for the snapshot

B. A Systems Manager Automation runbook with a wait step, used as the Config remediation action

C. An EventBridge rule that deletes the volume immediately

D. A cron job on an instance

◆ PAUSE AND DECIDE

What should implement the remediation?

A — A single Lambda function that waits for the snapshot

Lambda stops after 15 minutes; snapshots can take hours.

B — A Systems Manager Automation runbook with a wait step, used as the Config remediation action

Runbooks can wait for a resource state for hours, retry steps, and are Config's native remediation target.

C — An EventBridge rule that deletes the volume immediately

Deletes before the snapshot exists.

D — A cron job on an instance

A server to run, and not triggered by Config.

REVISIT MODULE 22

Long waits, retries, triggered by Config: an Automation runbook as the remediation.

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> A Lambda function in private subnets with no NAT gateway must read objects from S3 and send messages to SQS. Both calls time out.
>
> Permissions are correct, and the company will not add internet access.

Which steps fix both calls? (Choose TWO.)

A. An S3 gateway endpoint on the subnets' route tables

B. Increase the function timeout

C. Move the function into public subnets

D. Set reserved concurrency

E. An SQS interface endpoint with private DNS

◆ PAUSE AND DECIDE

Which steps fix both calls? (Choose TWO.)

A An S3 gateway endpoint on the subnets' route tables

A free private route to S3.

B Increase the function timeout

There is no route; calls just fail more slowly.

C Move the function into public subnets

Lambda gets no public IP there either.

D Set reserved concurrency

Unrelated to network paths.

E An SQS interface endpoint with private DNS

A private path to SQS, which has no gateway endpoint.

REVISIT MODULE 23

Two services, two private paths: a gateway endpoint for S3, an interface endpoint for SQS.

A CloudTrail trail in account A delivers logs to a bucket in a log archive account. The bucket policy allows the CloudTrail service principal to write.

Security wants to ensure only that specific trail can write, so no other account's trail could use the bucket.

What should the DevOps engineer add to the bucket policy?

A An sts:ExternalId condition

B A condition requiring MFA

C A condition on aws:SourceArn matching the trail's ARN

D A condition on aws:SourceIp

◆ PAUSE AND DECIDE

What should the DevOps engineer add to the bucket policy?

A An sts:ExternalId condition

External IDs are for third-party role assumption, not service principals.

B A condition requiring MFA

A service principal cannot use MFA.

C A condition on aws:SourceArn matching the trail's ARN

The service may write only on behalf of that trail: no confused deputy.

D A condition on aws:SourceIp

Service requests do not come from your addresses.

REVISIT MODULE 24

An AWS service acting for you: aws:SourceArn or aws:SourceAccount. A third party: ExternalId.

A new customer managed KMS key protects payroll data. Security requires that key administrators can manage the key but never decrypt, and that the payroll application can decrypt but never change the key.

Both groups work in the same account.

How should the key policy be written?

A Separate statements: management actions for the admin role, and Decrypt for the application role

B kms:* for both roles

C An AWS managed key instead

D Grant both roles through IAM policies only, with no key policy statements

◆ PAUSE AND DECIDE

How should the key policy be written?

A. Separate statements: management actions for the admin role, and Decrypt for the application role

Each principal gets only its own set of actions.

B. kms:* for both roles

Gives each group the other's powers.

C. An AWS managed key instead

Its policy cannot be customised at all.

D. Grant both roles through IAM policies only, with no key policy statements

Works only if the key policy enables IAM, and then IAM admins could grant themselves decrypt.

REVISIT MODULE 25

Separation of duties on a key: separate key policy statements for administrators and users.

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> Security wants to know which EC2 instances are reachable from the internet on any open port, and which of those also run vulnerable software.
>
> It wants this continuously, across many accounts, with no scanning agents to deploy beyond what is already there.

What should the DevOps engineer enable?

A VPC Flow Logs analysed with Athena

B Amazon Inspector, organisation-wide, for EC2 vulnerability and network reachability findings

C GuardDuty Runtime Monitoring

D Security groups reviewed monthly

◆ PAUSE AND DECIDE

What should the DevOps engineer enable?

A VPC Flow Logs analysed with Athena

Shows traffic that happened, not reachability or software.

B Amazon Inspector, organisation-wide, for EC2 vulnerability and network reachability findings

Reachability and vulnerabilities, combined and continuous.

C GuardDuty Runtime Monitoring

Detects active threats, not exposure or vulnerable packages.

D Security groups reviewed monthly

Manual, and ignores software.

REVISIT MODULE 27

Exposed and vulnerable: Inspector's reachability plus vulnerability findings.

> Auditors require that CloudTrail logs can be proven unaltered, and cannot be deleted by anyone for seven years. And no member account may stop the organisation trail.
>
> The company uses Organizations with a log archive account.

Which steps meet these requirements? (Choose THREE.)

A Enable CloudTrail log file validation

B Enable CloudTrail Insights

C S3 Object Lock in compliance mode with seven-year retention on the log bucket

D An SCP denying cloudtrail:StopLogging and DeleteTrail

E Amazon Macie on the log bucket

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose THREE.)

A Enable CloudTrail log file validation
Digest files prove whether logs were altered.

B Enable CloudTrail Insights
Flags unusual activity; proves nothing about integrity.

C S3 Object Lock in compliance mode with seven-year retention on the log bucket
Nobody, including root, can delete the logs early.

D An SCP denying cloudtrail:StopLogging and DeleteTrail
No member account can stop the trail.

E Amazon Macie on the log bucket
Finds sensitive data; protects nothing.

REVISIT MODULE 27

Proven, undeletable, unstoppable: validation, Object Lock, and an SCP.

Answer key

- **Answer key:** 1 C, 2 A, 3 D, 4 CE, 5 B, 6 C, 7 A, 8 D, 9 AD, 10 B
- **Answer key:** 11 A, 12 C, 13 BC, 14 D, 15 B, 16 AE, 17 C, 18 A, 19 B, 20 ACD
- Multiple response (4, 9, 13, 16, 20) score only if every letter is right.
- 15 or more: on track. 13 or 14: close. 12 or fewer: revisit weak domains first.

> ON THE EXAM
>
> Each solution slide named the module to revisit. Use it for every miss.

Scoring by domain

DOMAIN	QUESTIONS	MODULES TO REVISIT
D1 SDLC Automation	1, 2, 3, 4	2, 3, 5, 7
D2 Configuration Management and IaC	5, 6, 7	10, 25
D3 Resilient Cloud Solutions	8, 9, 10	15, 16
D4 Monitoring and Logging	11, 12, 13	17, 18
D5 Incident and Event Response	14, 15, 16	21, 22, 23
D6 Security and Compliance	17, 18, 19, 20	24, 25, 27

ON THE EXAM

Two misses in one domain matter more than three spread across six.